

Diophantine Equations

1. How many non-negative integer pairs (m, n) satisfy $5m + 7n = 503$?
2. A number is equal to three times the sum of its digits. Find the number.
3. How many ways are there to break a dollar into pennies, nickels, dimes and quarters?
4. Find all positive integer solutions (x, y) to

$$xy - 3x - 5y = 7.$$

5. How many lattice points (x, y) satisfy $|x| + |y| \leq 7$ and $xy > 0$?
6. How many ordered pairs (x, y) of integers satisfy $x^2 - y^2 = 105$?
7. Find the number of ordered pairs (x, y) of non-negative integers satisfying

$$3^x + 3^y = 3^{10}.$$

8. The product of three consecutive integers equals 8 times their sum. Find all solutions.
9. Find the number of ordered pairs (a, b) of positive integers such that

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{20}.$$

10. Find the number of ways to write 2025 as an ordered product of two positive integers (a, b) with $a < b$ such that $b - a$ is a perfect square.
11. Find the number of ordered triples (x, y, z) of positive integers satisfying

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{4}$$

with $x \leq y \leq z$.

12. Determine the number of ordered pairs (m, n) of positive integers such that $m^2 - n^2 = 2^{100}$.
13. Find the number of positive integer solutions (x, y) to

$$x^2 + y^2 = x^2y^2 - 2xy.$$

14. For how many integers n with $1 \leq n \leq 500$ does the equation

$$xy = n(x + y)$$

have at least one positive integer solution (x, y) ?

15. Find all triples (x, y, z) of positive integers with $x \leq y \leq z$ satisfying

$$xyz = 2(xy + yz + zx).$$

16. A positive integer n can be written as a difference of two squares in exactly 3 ways. What is the smallest such n ?

17. Find the number of ordered pairs (a, b) of positive integers with $a \leq b$ such that $\text{lcm}(a, b) = 2^3 \cdot 3^2 \cdot 5$.

18. The equation $x^3 - y^3 = 217$ has a solution in positive integers. Find $x + y$.

19. How many ordered pairs of non-negative integers (x, y) satisfy

$$x^2 + 3y^2 = 75?$$

20. Find the number of ordered pairs (x, y) of positive integers with $x < y$ such that

$$x + y + xy = 120.$$

21. Find the sum of all positive integers k such that

$$\frac{k}{36 - k}$$

is also a positive integer.

22. How many ordered pairs (x, y) of integers satisfy

$$(x^2 - 4)(y^2 - 9) = 72?$$

23. Find all positive integer solutions (x, y, z) to

$$x! + y! = z!.$$

24. Find the number of ordered triples of positive integers (a, b, c) with $a \leq b \leq c$ satisfying

$$a + b + c = abc.$$

25. How many ordered pairs (x, y) of positive integers satisfy

$$x^2y + xy^2 = 180?$$

26. Find the number of ordered pairs (m, n) of positive integers such that

$$m + n = \text{gcd}(m, n)^2$$

and $\text{gcd}(m, n) < 20$.

- 27.** Let p be a prime. For how many primes $p < 1000$ does the equation

$$x^2 + y^2 = p$$

have a solution in non-negative integers (x, y) ?

- 28.** Find all ordered pairs (x, y) of positive integers such that $x^y = y^x$ and $x \neq y$.

- 29.** Find all integer solutions (x, y) to

$$2^x = y^2 + 1.$$

- 30.** Find the number of ordered pairs (x, y) of integers satisfying

$$x^2 - xy + y^2 = 7.$$

- 31.** Let N be the number of ordered pairs of integers (x, y) satisfying

$$x^2 + y^2 \leq 25 \quad \text{and} \quad xy = 6.$$

Find N .

- 32.** Find the number of ordered pairs (a, b) of positive integers such that $a \cdot b = 10!$ and $\gcd(a, b) = 1$.

- 33.** Find all positive integer solutions to

$$1 + x + x^2 + x^3 = 2^y.$$

- 34.** Find the number of ordered pairs of positive integers (m, n) such that $m + n = 2024$ and $\gcd(m, n) = 8$.